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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

**“A LEARNING MODEL ON CUSTOMER CHURN
FORECASTING FOR TELECOM PROVIDERS”**

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Presented by – BATCH 42 (CSE - C)

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OVERVIEW

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ABSTRACT

In the competitive telecom industry, keeping customers is vital for profit and growth. This project aims to understand why customers leave their telecom providers. By looking at past data on customer behavior, service usage, and interactions, we want to find patterns that show when a customer might leave. We will collect and study different information, like account details, service history, and feedback. This will help us identify key signs of potential departure and the reasons behind it. With these insights, telecom companies can create strategies to improve customer satisfaction, enhance service quality, and take proactive steps to keep their customers. The main goal is to provide useful information that helps telecom providers reduce customer turnover, improve retention efforts, and build long-term relationships with their customers.

Algorithms

Cox Proportional Hazard Model:

A statistical method that examines the effect of multiple variables on the time to an event (e.g., customer churn). It calculates hazard ratios for predictors.

Literature review-1 **Applying Survival Analysis to** **Telecom Churn Data**

Key Findings

- Cox Proportional Hazard Model was applied to predict customer churn using various predictors like campaign effectiveness, tariff plans, tenure, and age.
- Tenure and auto-payment positively impact customer retention.
- Customers with shorter tenure and no auto-payment have higher churn probabilities



Model: Cox Proportional Hazard Model

A statistical method used to examine the effect of multiple variables on the time until an event (customer churn) occurs.

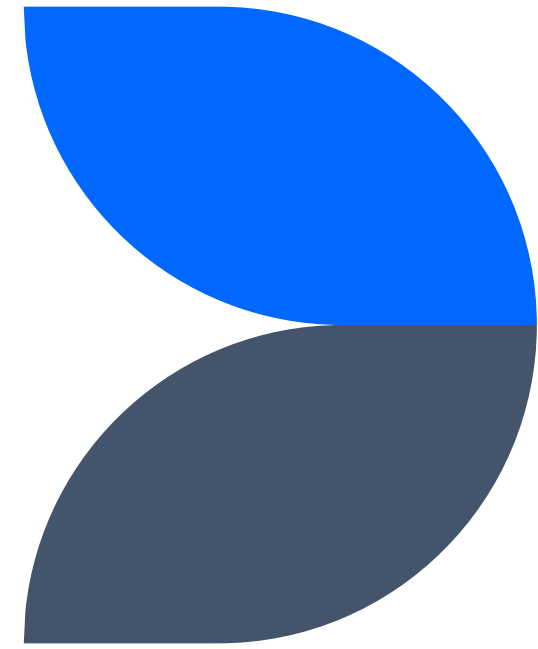
Focuses on survival and hazard functions to identify high-risk customers.

Merits

- Accurate prediction of churn probabilities over time using survival analysis.
- Identifies key factors influencing customer churn (e.g., tenure, auto-payment).

Demerits

- Considered the survival chances of consumers



Literature review-2

Customer Churn Prediction in Telecom Sector Using Machine Learning Techniques

Algorithms

Decision Tree: A tree-like structure where nodes represent features, and branches represent decision outcomes, making it interpretable and easy to implement.

Random Forest: An ensemble learning method using multiple decision trees and averaging their results to improve accuracy and reduce overfitting

Key Findings

- **Accuracy:** Achieved a prediction accuracy of 99% using the Random Forest classifier.
- **Precision and Recall:** The classifier matrix showed 99% precision and recall, with an overall accuracy of 99.09%.
- **Provided an efficient business model for churn prediction to enable proactive actions by identifying churners early.**



Customer Churn Prediction in Telecom Sector Using Machine Learning Techniques

Model: Random Forest Classifier

An ensemble learning model that builds multiple decision trees and combines their outputs for improved accuracy and reduced overfitting.

Merits

- High accuracy (99.09% using Random Forest).
- Easy implementation and scalability.

Demerits

- May overfit on small datasets.
- Requires balanced datasets for optimal results.

Literature review-3

Customer Churn Prediction in Telecom Using Machine Learning in Big Data Platform

Algorithms

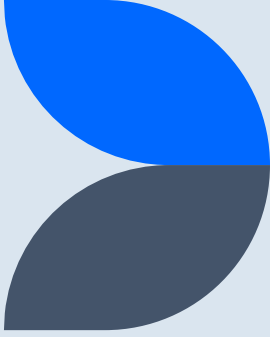
XGBoost: An advanced gradient boosting algorithm that uses decision trees. It is efficient, handles missing values, and offers high predictive performance.

Gradient Boosted Machines (GBM): A machine learning technique that builds models sequentially, where each model corrects errors from the previous one.

Key Findings

- Achieved an AUC of 93.3% using the XGBoost algorithm with Social Network Analysis (SNA) features.
- Improved model performance significantly (from 84% to 93.3% AUC).
- XGBoost outperformed Gradient Boost Machines, Random Forest, and Decision Trees in terms of prediction accuracy and scalability.

Customer Churn Prediction in Telecom Using Machine Learning in Big Data Platform



Model: XGBoost (Extreme Gradient Boosting)

A gradient boosting framework optimized for speed and performance, which builds decision trees sequentially to minimize prediction errors.

Merits

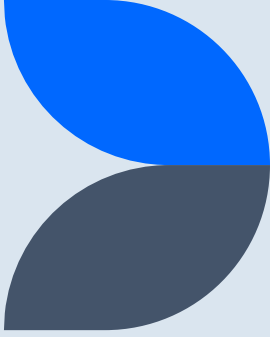
- High AUC score (93.3% using XGBoost).
- Scalable to large datasets (70TB).

Demerits

- High computational cost for SNA.
- Complexity in feature engineering.

Literature review -4

Customer Churn Prediction System: A Machine Learning Approach



Algorithms

Gravitational Search Algorithm (GSA): An optimization technique inspired by Newtonian gravity that selects relevant features by simulating interactions between objects in a search space.

Support Vector Machine (SVM):

A supervised learning algorithm that finds the hyperplane that best separates classes.

Key Findings

- AdaBoost and XGBoost classifiers achieved the highest accuracy of 81.71% and 80.8%.
- The Gravitational Search Algorithm (GSA) effectively reduced data dimensionality, improving overall performance.
- Demonstrated the value of ensemble learning and optimization techniques in improving prediction accuracy.

Customer Churn Prediction System: A Machine Learning Approach

Model: AdaBoost and XGBoost Classifiers

AdaBoost: Boosting model that combines weak classifiers iteratively.

XGBoost: Advanced gradient boosting algorithm focusing on efficiency.

Merits

- Reduces data dimensionality and prevents overfitting.
- AUC of 84% using AdaBoost and XGBoost.

Demerits

- Limited improvement in class imbalance.
- High dependency on feature selection accuracy.

Literature review-5

The perils of proactive churn prevention- Using plan recommendations

Algorithms

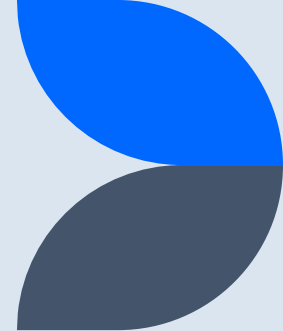
Randomized Control Trial (RCT): Used to evaluate the impact of encouraging customers to switch to cost-minimizing plans.

Behavioral Analysis: Applied to understand the unintended effects of interventions, such as lowering customer inertia and increasing the salience of past usage patterns.

Key Findings

- Encouraging customers to switch to cost-minimizing plans unexpectedly increased churn rates.
- Churn in the treatment group was 10%, compared to 6% in the control group.
- Salience of Past Usage Patterns: The encouragement highlighted overspending or dissatisfaction, triggering latent churn behavior.





The perils of proactive churn

Using prevention-plan recommendations

Model: Randomized Control Trial (RCT) and Behavioral Analysis

Behavioral Model: Explores the psychological and behavioral effects of interventions on customer decision-making.

Merits

- Highlights potential unintended consequences of proactive churn prevention strategies.

Demerits

- No machine learning or automated prediction models were used.

Literature review-6

Customer churn prediction using machine learning- A study in the B2B subscription

Algorithms

XGBoost: Decision-tree-based ensemble boosting technique for strong classifier performance.

Random Forest: An ensemble learning method using multiple decision trees and averaging their results to improve accuracy and reduce overfitting.

Key Findings

XGBoost and Random Forest outperformed Naïve Bayes in predicting churn, with higher accuracy, precision, and recall.

Metrics such as accuracy, precision, recall, and F1-score were used to assess the models. Ensemble methods performed consistently better than single models like Naïve Bayes.

Customer churn prediction using machine learning- A study in the B2B subscription

Model: Machine Learning-based Churn Prediction

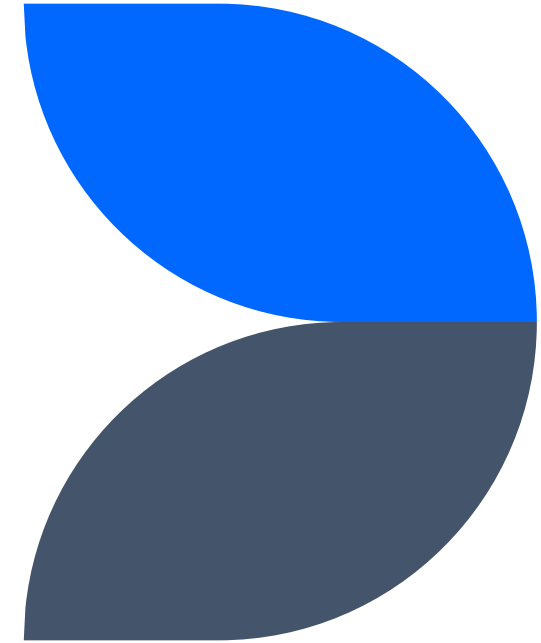
A binary classification model to predict whether a customer will churn in a subscription-based B2B context.

Merits

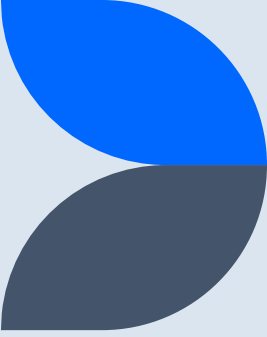
- Sampling methods improved classifier performance on imbalanced data.
- Comprehensive evaluation using metrics like accuracy, precision and recall.

Demerits

- Limited focus on economic implications of churn prediction.
- Study specific to financial administration in the B2B context.

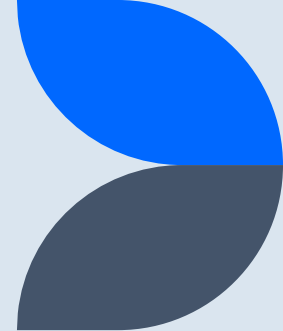


PROPOSED SYSTEM



Algorithms

- **Logistic Regression:** A statistical algorithm used for binary classification tasks by modeling the probability of a certain class using a sigmoid function.
- **Support Vector Classifier:** A supervised learning algorithm that identifies the hyperplane maximizing the margin between data classes.
- **Naive Bayes:** A probabilistic algorithm based on Bayes' theorem, assuming strong independence between features.
- **Decision Tree:** A tree-like model where nodes represent feature splits and leaves represent class labels, making it interpretable.
- **Random Forest:** An ensemble algorithm that combines multiple decision trees using bagging to reduce overfitting and improve accuracy.



Data-Driven Model:

Gather relevant data such as customer demographics, usage patterns, and churn history.

Enhanced Machine Learning Approach:

Perform model evaluation using metrics such as Accuracy, Precision, Recall, and F2 Score.

Advanced Optimization Techniques:

Handle class imbalance using resampling techniques like SMOTE.

Merits:

- Improved Feature Selection and Handling Class Imbalance
- Dynamic and Adaptive Model Updates

IMPLEMENTATION

- We used python to implement this project.
- Python is a widely used general-purpose, high level programming language. It was created by Guido van Rossum in 1991 and further developed by the Python Software Foundation.
- **Advantages of python**
 - Extensive Libraries
 - Simple and Easy
 - Readable
 - Object-Oriented
 - Free and Open-Source
 - Portable

IMPLEMENTATION

- Machine learning (ML) is a type of artificial intelligence (AI) that allows software applications to become more accurate at predicting outcomes without being explicitly programmed to do so.
- **Types of Machine Learning**
 - Supervised learning
 - Unsupervised learning
 - Semi-supervised learning
 - Reinforcement learning

IMPLEMENTATION

```
#Standard libraries for data analysis:-----
```

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
from scipy.stats import norm, skew
from scipy import stats
import statsmodels.api as sm
```

```
#Standard libraries for data visualization-----
```

```
import seaborn as sn
from matplotlib import pyplot
import matplotlib.pyplot as plt
import matplotlib.pylab as pylab
import matplotlib
%matplotlib inline
color = sn.color_palette()
import matplotlib.ticker as mtick
from IPython.display import display
pd.options.display.max_columns = None
from pandas.plotting import scatter_matrix
```

```
# Clean the Dataset-----
```

```
dataset['TotalCharges'] = pd.to_numeric(dataset['TotalCharges'],errors='coerce')
dataset['TotalCharges'] = dataset['TotalCharges'].astype("float")
```

```
# Take care of missing data-----
```

```
dataset.info
```

IMPLEMENTATION

```
##Find the average and fill missing values of each columns programmatically.
na_cols = dataset.isna().any()

na_cols = na_cols[na_cols == True].reset_index()

na_cols = na_cols["index"].tolist()

for col in dataset.columns[1:]:
    if col in na_cols:
        if dataset[col].dtype != 'object':
            dataset[col] = dataset[col].fillna(dataset[col].mean()).round(0)
```

```
#Histogram:

fig = plt.figure(figsize=(15, 12))
plt.suptitle('Histograms of Numerical Columns\n',horizontalalignment="center",fontstyle = "normal", fontsize = 12)
for i in range(dataset2.shape[1]):
    plt.subplot(6, 3, i + 1)
    f = plt.gca()
    f.set_title(dataset2.columns.values[i])

    vals = np.size(dataset2.iloc[:, i].unique())
    if vals >= 100:
        vals = 100

    plt.hist(dataset2.iloc[:, i], bins=vals, color = '#1b8f90')
plt.tight_layout(rect=[0, 0.03, 1, 0.95])
```

IMPLEMENTATION

```
#(1) Distribution of Contract Type-----
contract_split = dataset[["customerID", "Contract"]]
sectors = contract_split.groupby("Contract")
contract_split = pd.DataFrame(sectors["customerID"].count())
contract_split.rename(columns={'customerID': 'No. of customers'}, inplace=True)

ax = contract_split[["No. of customers"]].plot.bar(title = 'Customers by Contract Type', legend = True, table = False, grid = False, subplots = False, figsize =(10, 5), color = '#aa8ffa', fontsize = 12, stacked=False)

plt.ylabel('No. of Customers\n',horizontalalignment="center",)
plt.xlabel('\n Contract Type',horizontalalignment="center")
plt.title('Customers by Contract Type \n',horizontalalignment="center")
plt.legend(loc='upper right', fontsize = "medium")
plt.xticks(rotation=0, horizontalalignment="center")
plt.yticks(rotation=0, horizontalalignment="right")

x_labels = np.array(contract_split[["No. of customers"]])

def add_value_labels(ax, spacing=5):
    for rect in ax.patches:
        y_value = rect.get_height()
        x_value = rect.get_x() + rect.get_width() / 2
        space = spacing
        va = 'bottom'
        if y_value < 0:
            space *= -1
            va = 'top'
        label = "{:.0f}".format(y_value)
        ax.annotate(
            label,
            (x_value, y_value),
            xytext=(0, space),
            textcoords="offset points",
            ha='center',
            va=va)
    add_value_labels(ax)
```

IMPLEMENTATION

```
# Compare Baseline Classification Algorithms - First Iteration
#Using Accuracy and ROC AUC Mean Metrics
from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC
from sklearn.neighbors import KNeighborsClassifier
from sklearn.naive_bayes import GaussianNB
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier

models = []

models.append(('Logistic Regression', LogisticRegression(solver='liblinear', random_state = 0, class_weight='balanced')))

models.append(('SVC', SVC(kernel = 'linear', random_state = 0)))

models.append(('Gaussian NB', GaussianNB()))

models.append(('Decision Tree Classifier', DecisionTreeClassifier(criterion = 'entropy', random_state = 0)))

models.append(('Random Forest', RandomForestClassifier(n_estimators=100, criterion = 'entropy', random_state = 0)))

#Evaluating Model Results:

acc_results = []
auc_results = []
names = []
# set table to table to populate with performance results
col = ['Algorithm', 'ROC AUC Mean', 'ROC AUC STD', 'Accuracy Mean', 'Accuracy STD']

model_results = pd.DataFrame({"Algorithm":pd.Series(dtype='str'),
                              "ROC AUC Mean" : pd.Series (dtype='float'),
                              "ROC AUC STD" : pd.Series (dtype='float'),
                              "Accuracy Mean" : pd.Series (dtype='float'),
                              "Accuracy STD" : pd.Series (dtype='float')})

#, "ROC AUC STD", "Accuracy Mean", "Accuracy STD"])
```

IMPLEMENTATION

```
# Evaluate the model using ROC Graph
from sklearn.metrics import auc, roc_auc_score, roc_curve
classifier.fit(X_train, y_train)
probs = classifier.predict_proba(X_test)
probs = probs[:, 1]
classifier_roc_auc = accuracy_score(y_test, y_pred )

rf_fpr, rf_tpr, rf_thresholds = roc_curve(y_test, classifier.predict_proba(X_test)[:,1])
plt.figure(figsize=(10, 5))

# Plot Logistic Regression ROC
plt.plot(rf_fpr, rf_tpr, label='Logistic Regression (area = %0.2f)' % classifier_roc_auc)
# Plot Base Rate ROC
plt.plot([0,1], [0,1],label='Base Rate' 'k--')

plt.xlim([0.0, 1.0])
plt.ylim([0.0, 1.05])

plt.ylabel('True Positive Rate \n',horizontalalignment="center")
plt.xlabel('\nFalse Positive Rate \n',horizontalalignment="center")
plt.title('ROC Graph \n',horizontalalignment="center")
plt.legend(loc="lower right", fontsize = "medium")
plt.xticks(rotation=0, horizontalalignment="center")
plt.yticks(rotation=0, horizontalalignment="right")
plt.show()
```


RESULTS

	Algorithm	ROC AUC Mean	ROC AUC STD	Accuracy Mean	Accuracy STD
0	Logistic Regression	84.12	1.65	74.60	1.26
1	SVC	83.64	1.68	79.98	1.08
2	Gaussian NB	81.82	1.79	68.99	1.46
4	Random Forest	81.72	2.02	78.47	1.57
3	Decision Tree Classifier	65.44	1.67	72.75	1.45

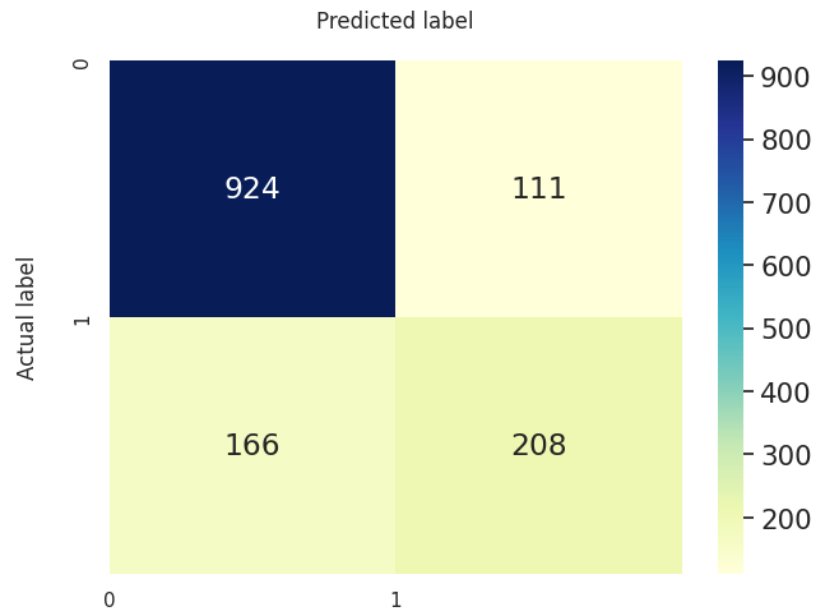
Baseline Classification Algorithms - Using Accuracy and ROC AUC Mean Metrics

	Model	Accuracy	Precision	Recall	F1 Score	F2 Score
0	Logistic Regression	0.803407	0.652038	0.556150	0.600289	0.573003
1	SVM (Linear)	0.803407	0.650155	0.561497	0.602582	0.577240
4	Random Forest	0.777857	0.611722	0.446524	0.516229	0.472018
3	Decision Tree	0.739532	0.508997	0.529412	0.519004	0.525199
2	Naive Byes	0.703336	0.467359	0.842246	0.601145	0.725806

Visualize the results and compare the baseline algorithms on Test data

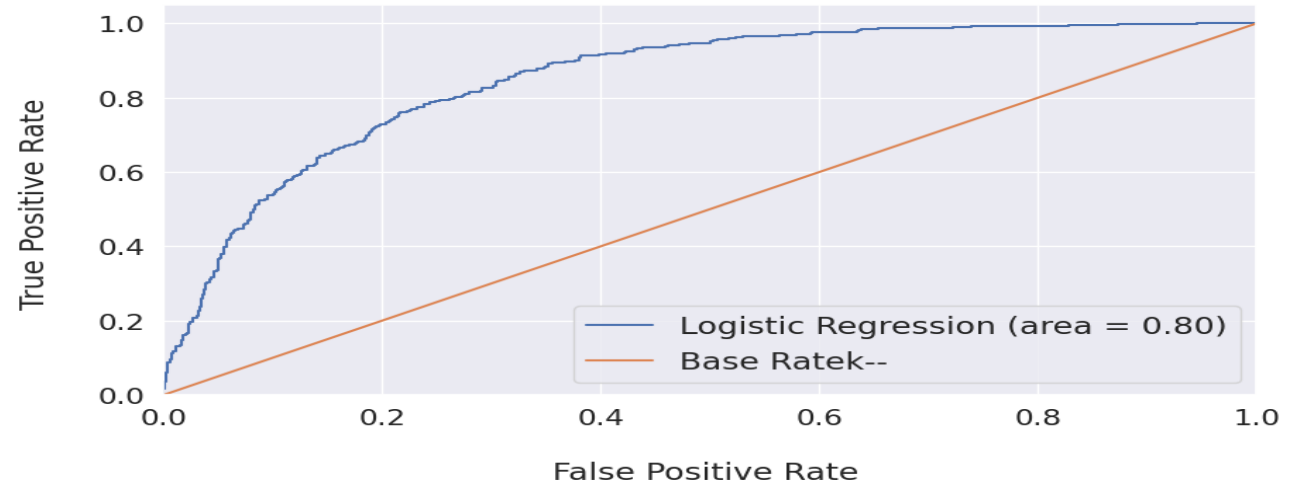
RESULTS

Confusion matrix



Confusion Matrix for Logistic Regression

ROC Graph



ROC Graph for Logistic Regression

RESULTS

	Model	Accuracy	Precision	Recall	F1 Score	F2 Score
3	Decision Tree	0.997946	0.999758	0.996134	0.997943	0.996857
4	Random Forest	0.997946	0.998067	0.997826	0.997946	0.997874
0	Logistic Regression	0.776033	0.761024	0.804784	0.782292	0.795634
1	SVM (Linear)	0.746920	0.707302	0.842474	0.768993	0.811459
2	Naive Byes	0.748128	0.704337	0.855279	0.772504	0.820128

Baseline Classification Algorithms - Using Accuracy and ROC AUC Mean Metrics after performing SMOTE oversampling technique.

	Model	Accuracy	Precision	Recall	F1 Score	F2 Score
4	Random Forest	0.803865	0.837809	0.753623	0.793489	0.769079
3	Decision Tree	0.747343	0.778261	0.691787	0.732481	0.707510
0	Logistic Regression	0.787923	0.769928	0.821256	0.794764	0.810450
2	Naive Byes	0.761836	0.718196	0.861836	0.783487	0.828688
1	SVM (Linear)	0.745894	0.709810	0.831884	0.766014	0.804222

Visualize the results and compare the baseline algorithms on Test data



CONCLUSION

For modern subscription-based businesses, managing customer churn, and specifically predicting churn, is crucial. Hence, the aim of this study was to develop a system capable of predicting customer churn in business. Our findings indicate that machine learning can predict customer attrition with high accuracy. Through machine learning, we explored various aspects related to customer churn prediction, contributing to a better understanding of how machine learning can be applied in this context. In this paper, we tested five classification algorithms on a customer dataset to predict customer churn. Our model pipeline includes several data preprocessing steps, such as visualizing data, removing features based on correlation and VIF, and addressing class imbalance. The set of records is split between tests and training information in an 80:20 ratio, and the data is normalized before applying machine learning algorithms. The classification algorithms used in the study are Support Vector Classifier, Logistic Regression, Gaussian Naïve Bayes, Decision Tree Classifier, Random Forest. The performance of these algorithms is evaluated using various metrics.



THANK YOU